

Fraol Batole

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PhD Student — Software Engineering for AI | Coding Agents | Program Analysis

EDUCATION

- Tulane University — Ph.D., Computer Science** Aug 2024 – Present
New Orleans, Louisiana | GPA: 4.0/4.0 | Advisor: Prof. Hridesh Rajan
- Iowa State University — M.Sc., Computer Science** May 2024
Ames, Iowa | GPA: 3.62/4.0
- Addis Ababa Science and Technology University — B.Sc., Software Engineering** June 2018
Addis Ababa, Ethiopia | GPA: 3.75/4.0 | Graduated Magna Cum Laude | Founded Innovation Club

RESEARCH EXPERIENCE

- Laboratory for Software Engineering, Tulane University** Aug 2024 – Present
Graduate Research Assistant New Orleans, Louisiana
- **LLM Agent Reliability (RefineAct):** Built a runtime verification framework that automatically derives task-specific formal specifications and enforces them during live LLM agent execution — detects failure modes invisible to static analysis or prompt engineering alone. *ASE '26*.
 - **Agent-Oriented Design Issue Localization (LocalizeAgent):** Built a multi-agent framework integrating static analysis to produce abstraction-aware, LLM-friendly code summaries for automated design-issue localization. *ICSE '25*.
 - **Typestate-Based Fault Localization:** Developed static analysis techniques to detect API usage violations in deep learning programs using typestate analysis. *TSE '26*.
 - **LLM Agent Bug Study:** Led the first large-scale empirical study of LLM-based agent bugs, analyzing 1,016 bug reports and identifying tool integration (45%) and model integration (16%) as primary failure modes (under submission).
 - **Move Method Refactoring (MM-Assist): JetBrains Collaboration:** Developed and released MM-Assist, an IntelliJ IDEA plugin for automated Move Method refactoring using LLMs, RAG, and hallucination filtering — achieving 73% accuracy and 4× improvement over prior tools (ICSME '25).
- Polytechnique Montréal — Research Intern** May 2025 – Aug 2025
Supervisor: Prof. Foutse Khomh Montreal, Canada
- Investigated the safety and robustness of LLM-based AI agents, focusing on novel detection and mitigation strategies for agent misbehavior. The work directly informs RefineAct.

TECHNICAL SKILLS

Programming Languages: Python, Java
ML / LLM Frameworks: PyTorch, TensorFlow/Keras, Hugging Face Transformers, fine-tuning, RAG pipelines
Agents & Tooling: LangChain, LangGraph, CrewAI, multi-agent frameworks, Model Context Protocol (MCP)
Program Analysis / SE: Static analysis, AST manipulation, typestate analysis, runtime verification, JetBrains Platform SDK (IntelliJ plugin development)
Cloud / Infrastructure: Google Cloud Platform, distributed training, HPC clusters (ACCESS/XSEDE)

SELECTED PUBLICATIONS (FIRST-AUTHOR)

Accepted

1. **Fraol Batole**, Foutse Khomh, and Hridesh Rajan. "Automatic Runtime Verification of LLM Agent Actions" *ASE'26*
2. **Fraol Batole**, Ruchira Manke, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "Typestate-Based Fault Localization of API Usage Violations in Deep Learning Programs." *IEEE Transactions on Software Engineering (TSE '26)*.
3. **Fraol Batole**, David O'Brien, Tien N. Nguyen, Robert Dyer, and Hridesh Rajan. "An LLM-Based Agent-Oriented Approach for Automated Code Design Issue Localization." *ICSE '25*.

Under Submission

4. **Fraol Batole**, Smit Patel, Aashish Yadavally, Tien N. Nguyen, and Hridesh Rajan. "CAPS: Code Abstraction Pre-training Strategy for Large Language Models."
5. **Fraol Batole**, Ruchira Manke, David O'Brien, Steven Vo, Tien N. Nguyen, and Hridesh Rajan. "A Comprehensive Study on LLM-Based Agent Bug Characteristics."

Full Publications & Co-authored Work

Accepted

6. Abhiram Bellur, Mohammed Raihan Ullah, **Fraol Batole** et al. "Multi-Agent Coordinated Rename Refactoring." FSE'26 (Tool Demo Track)
7. Abhiram Bellur, **Fraol Batole** et al. "LLM, IDE and Semantic Embedding to Assist Move Method Refactoring." ICSME '25.
8. Sayem Imtiaz, Astha Singh, **Fraol Batole**, and Hridesh Rajan. "IRepair: An Intent-Aware Approach to Repair Data-Driven Errors in LLMs." FSE '25.
9. Sayem Imtiaz, **Fraol Batole** et al. "Decomposing RNN to Enable Reuse." ICSE '23.
10. Abhiram Bellur, **Fraol Batole**. "IDE Native, Foundation Model-Based Agents for Software Refactoring." IDE Workshop @ ICSE '25.

Under Submission

11. Samuel W. Flint, Robert Dyer, **Fraol Batole**, and David O'Brien. "Do LLMs Generate Code with Type Annotations?"
12. David O'Brien, **Fraol Batole** et al. "Issue-Based Programming Language Evolution."

ACADEMIC SERVICE & AWARDS

Distinguished Reviewer Award (Junior PC) — MSR '26

April 2026

Reviewer / Program Committee:

- Junior Program Committee (Technical Track) — Mining Software Repositories (MSR '26)
- Reviewer — Transactions on Software Engineering and Methodology (TOSEM '26)
- Reviewer — Automated Software Engineering (ASE) Journal '25
- Reviewer — Deep Learning for Code Workshop @ NeurIPS '25

ADDITIONAL EXPERIENCE

Iowa State University — Graduate Teaching Assistant

Jan 2021 – Aug 2024

CS Department

Ames, Iowa

- Taught programming principles, debugging, and test-case development to 80+ students. Adapted pedagogy for diverse learning styles.

MED Innovation — Software Engineer

Aug 2019 – Nov 2019

Addis Ababa, Ethiopia

- Integrated a CRM system for a cross-functional team using Spring Boot and Vaadin.

HONORS & GRANTS

- **NSF-Funded Advanced Computing Credits:** 1.5 million credits awarded (Feb 2024, Jun 2025).
- **Google Cloud Research Credits:** \$12,000 awarded (Jun 2023).
- **Google Cloud Research Innovator '23:** Competitively selected for Google's research program (Mar 2023).
- **Magna Cum Laude, B.Sc. Software Engineering** — Addis Ababa Science and Technology University.